

1 Topological Spaces

Definition 1.0.1. Let X be any set. A topology T on X is a collection of subsets of X , which satisfies the following axioms.

- $\emptyset$ and X are both in T
- Closed under arbitrary unions
- Closed under finite intersections

The sets in T are called open sets.

A set X equipped w/ a topology is called a topological space.

Example 1.0.2.

- X is any set, $T = P(X)$. (discrete topology)
- X is any set, $T = \{\emptyset, X\}$. (indiscrete topology)
- $X = \{a, b, c\}$, $T = \{\emptyset, X, \{a, b\}\}$
- $X = \mathbb{R}$, $T = \{\emptyset, X\} \cup \{U \subset \mathbb{R} \mid \text{for all } x \in U, \exists \epsilon > 0 \text{ s.t. } (x - \epsilon, x + \epsilon) \subset U\}$

Proposition 1.0.3. *This is a topology.*

Proof.

- $\emptyset, \mathbb{R}$ are both open
- Suppose $\{U_j\}_{j \in J}$ is a collection of open sets in $\mathbb{R}$
 Let $x \in \bigcup_{j \in J} U_j$. Then $x \in U_j$ for some $j \in J$.
 U_j is open thus there is an open interval in U_j containing x . So it is in the arbitrary union
- Let U, V be open subsets of $\mathbb{R}$
 Let $x \in U \cap V$. Because U, V open there are intervals $(a, b) \subset U$, $(c, d) \subset V$ s.t.
 $x \in (a, b) \cap (c, d)$
 $(a, b) \cap (c, d) = (\max(a, c), \min(b, d)) \subset U \cap V$. Thus, x is in this interval thus
 $U \cap V$ is open.

□

- X is any set, let T be the collection of subsets $U \subseteq X$ s.t. either $U = \emptyset$, or $X \setminus U$ is finite.

Proposition 1.0.4. *This is a topology. (cofinite topology)*

Proof.

- The empty set is open, and $X \setminus X = \emptyset$ is finite, so $X \in T$
- Consider sets $\{U_j\}_{j \in J}$ s.t. $U_j \in T$
 First: consider the case for some j' , $X \setminus U_{j'}$ is finite.
 $X \setminus \bigcup_{j \in J} U_j = \bigcap_{j \in J} X \setminus U_j \subset X \setminus U_{j'}$. So the arbitrary union is finite.
 Second: If the first case does not occur, then $U_j = \emptyset$ for all $j \in J$, so the union is the empty set.
- Let $U, V \in T$. If either is the empty set then $U \cap V = \emptyset \in T$, so assume that both are not empty. Then consider $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$, so it is a union of two finite sets which is finite thus $U \cap V \in T$

□

Definition 1.0.5. If X is a set with topology T , then the complement of an open set in X is called a closed set.

Example 1.0.6. If T is cofinite topology, a set in $A \subseteq X$ is closed $\iff$ either finite or $A = X$

IMPORTANT: Some sets are neither open nor closed. Some sets are both closed and open.

1.1 Basis for a topology

Definition 1.1.1. Let X be a set. A collection β of subsets of X is a basis for a topology if:

- every $x \in X$ is in some $B \in \beta$
- If $x \in B_1 \cap B_2$ for $B_1, B_2 \in \beta$
then: $\exists B_3 \in \beta$ s.t. $x \in B_3 \subseteq B_1 \cap B_2$

The topology generated by β is the collection T of subsets defined by:

$$U \in T \iff \text{for every } x \in U, \exists B \in \beta \text{ s.t. } x \in B \subseteq U$$

Example 1.1.2.

$X = \mathbb{R}$, β = set of open intervals in $\mathbb{R}$

$X = \mathbb{R}^n$, β = set of open balls $B(x, r) = \{y \in \mathbb{R}^n \mid d(x, y) < r\}$

Check that the second example is actually a basis for a topology

- every $x \in \mathbb{R}^n$ lies in some open ball. $x \in B(x, 1)$
- Assume $x \in B_1 \cap B_2 = B(y_1, r_1) \cap B(y_2, r_2)$
Set $r_3 = \min(r_1 - d(x, y_1), r_2 - d(x, y_2))$
Consider the ball $B_3 = B(x, r_3)$. Let $z \in B_3$.
 $d(z, y_1) \leq d(x, y_1) + d(z, x) < d(x, y_1) + r_3 \leq d(x, y_1) + r_1 - d(x, y_1) = r_1$
So $z \in B(y_1, r_1) = B_1$ do the same proof for B_2 thus $z \in B_1 \cap B_2$ thus $B_3 \subseteq B_1 \cap B_2$

Proposition 1.1.3. This is actually a topology on X .

Proof.

- $\emptyset \in T$ vacuously, and $X \in T$ comes from first basis axiom.
- Suppose $\{u_j\}_{j \in J}$ of sets in T .
Let $x \in \bigcup_{j \in J} u_j$. Thus $x \in u_{j'}$ for some $j' \in J$, so $\exists B_{j'} \in \beta$ s.t. $x \in B_{j'} \subseteq u_{j'}$ thus $x \in B_{j'} \subseteq \bigcup_{j \in J} u_j$. This shows $\bigcup_{j \in J} u_j \in T$
- Suppose $x \in u_1 \cap u_2$ for $u_1, u_2 \in T$. Then $x \in B_1, B_2$ for sets $B_1, B_2 \in \beta$ s.t. $B_1 \subseteq u_1$ and $B_2 \subseteq u_2$ thus $x \in B_1 \cap B_2$. By the second axiom thus implies $\exists B_3 \in \beta$. Thus $x \in B_3 \subseteq B_1 \cap B_2 \subseteq u_1 \cap u_2 \in T$

□

Remark 1.1.4. Every topology is a basis for itself.

Proposition 1.1.5. Let X be a set, let β be a basis for topology T . The sets in T are precisely the sets of the form

$$\bigcup_{j \in J} B_j$$

for $\{B_j\}_{j \in J}$ collection of sets in β

Proof.

- Every set of this form is in T :

$$x \in \bigcup_{j \in J} B_j, \text{ then } x \in B_{j'} \subset \bigcup_{j \in J} B_j$$

- Every set in T has the form:

$$\text{Let } U \in T. \text{ Then for every } x \in U, \text{ there is a } B_x \in \beta \text{ s.t. } B_x \subset U \implies U = \bigcup_{x \in U} B_x$$

□

Decomposition is NOT in general unique.

Basis sets are always open.

1.1.1 Comparing Topologies

Definition 1.1.6. If T_1, T_2 are topologies on the same set X , we say that T_1 is finer than T_2 if $T_1 \supseteq T_2$. Also, T_2 is coarser than T_1

Proposition 1.1.7. Let β, β' be bases for topologies T, T' on X . Then $T \subseteq T'$ (T' is finer than T) if and only if every $x \in X$, and every $B \in \beta$ containing x , $\exists$ some $B' \in \beta'$ s.t. $x \in B' \subset B$

Example 1.1.8. $X = \mathbb{R}^n$, let β be the set of open boxes in $\mathbb{R}^n$: sets of the form $(a_1, b_1) \times (a_2, b_2) \times \cdots \times (a_n, b_n)$

This is a basis. It generates the same standard topology on $\mathbb{R}^n$.

Example 1.1.9. Let $X = \mathbb{R}$. Consider collections of half-open intervals in $\mathbb{R}$: $[a, b), (a, b]$.

Claim: this is a basis for a topology.

This is not the same as the std. topology on $\mathbb{R}$

This is finer than the std. topology.

1.2 Examples of topological spaces

Example 1.2.1 (Order Topology).

Let X be a set with the order relation $<$. For $a < b$ in X , define intervals:

$$(a, b) = \{x \in X \mid a < x < b\}$$

$$[a, b) = \{x \in X \mid a \leq x < b\}$$

Similar for $(a, b], [a, b]$

Let β be the set of all intervals in X of the forms:

- (a, b) for $a < b$ in X .
- $[a_0, b)$ where a_0 is the smallest element of X (if it exists).
- $(a, b_0]$ where b_0 is the largest element of X (if it exists).

Proposition 1.2.2. *This is a basis for a topology on X .*

This is called the order topology on X .

Example 1.2.3. Standard topology on $\mathbb{R}$ is order topology, on $\mathbb{R}$

Example 1.2.4 (Product Topology). Let X, Y be topological spaces.

Let β be a collection of sets:

$$\{U \times V \mid U \text{ open in } X, V \text{ open in } Y\}$$

Proposition 1.2.5. *This is a basis for a topology.*

This is called the product topology on $X \times Y$

Example 1.2.6. $\mathbb{R} \times \mathbb{R}$ with product topology is std topology on $\mathbb{R}^2$

If β_1 is a basis for topology on X , and β_2 is a basis for topology on Y , then

$$\{B \times C \mid B \in \beta_1, C \in \beta_2\}$$

is a basis for the product topology.

This definition is valid for products of pairs of sets.

Lemma 1.2.7. Suppose X is a topological space. Let C be a collection of open subsets satisfying:

- for each open set $U \subseteq X$ and each $x \in U$, $\exists c \in C$ s.t. $x \in c \subseteq U$

Then, C is a basis for top. on X .

Proof. Check C is a basis.

- Every $x \in X$ in some $c \in C$. X is open, so there exists c s.t. $x \in c \subseteq X$
- If $c_1, c_2 \in C$, then for any $x \in c_1 \cap c_2$, $\exists c_3 \in C$ s.t. $x \in c_3 \subseteq c_1 \cap c_2$.
Follows from the fact that c_1, c_2 are open which implies $c_1 \cap c_2$ is open so can find c_3

Check: C generates same topology T on X .

First, suppose U is open in T . By assumption for all $x \in U$, there exists $c \in C$ s.t. $x \in c \subseteq U$, so U is open in the topology generated by C .

If U is open in topology generated by C . Then U is a union of sets in C . But every $c \in C$ is open in T , so union U is also open in T . $\square$

Example 1.2.8 (Metric Topology).

Definition 1.2.9. Let X be a set. A function $d : X \times X \rightarrow \mathbb{R}$ is a metric if it satisfies:

- $d(x, y) = 0 \iff x = y$
- $d(x, y) = d(y, x) \quad \forall x, y \in X$
- $d(x, y) + d(y, z) \geq d(x, z) \quad \forall x, y, z \in X$

A set X equipped with a metric d is a metric space.

Let β be the set of all metric balls:

$$B_d(x, r) = \{y \in X \mid d(x, y) < r\} \quad \text{for } x \in X, r > 0$$

Proposition 1.2.10. This is a basis for a topology

This is called metric topology

Definition 1.2.11. A topological space X is called metrizable if $\exists$ metric d on X inducing topology on X .

Remark 1.2.12. There could be more than one metric inducing the same topology.

Example 1.2.13 (Subspace Topology). Let X be a set with topology $\mathcal{T}$. Let $Y \subseteq X$. Thus subspace topology on Y is the collection of sets:

$$\{U \cap Y \mid U \in \mathcal{T}\}$$

Proposition 1.2.14. *This is a topology on Y .*

We call Y a subspace of topology X .

Example 1.2.15. Suppose (X, d) is a metric space. $Y \subseteq X$.

The function d is also a metric on Y . We have 2 topologies on Y :

- Subspace topology as a subspace of X with metric topology.
- Metric topology on Y

Proposition 1.2.16. *These two topologies are the same.*

Proof. Let $\mathcal{T}_{\subseteq}$ is subspace topology, and $\mathcal{T}_d$ is metric topology on Y .

$$\mathcal{T}_d \text{ has a basis } \beta_d = \{B_d(y, r) \cap Y \mid y \in Y, r > 0\}$$

$$\mathcal{T}_{\subseteq} \text{ has a basis } \beta_{\subseteq} = \{B_d(x, r) \cap Y \mid x \in X, r > 0\}$$

Subspace Basis is just X 's basis but intersected with Y .

Easy to see that $\beta_d \subseteq \beta_{\subseteq}$ thus $\mathcal{T}_d \subseteq \mathcal{T}_{\subseteq}$

Let $y \in B \in \beta_{\subseteq}$, so $y \in B_d(x, r) \cap Y$ for some $x \in X, r > 0$

Using the triangle inequality, find some $r' > 0$ s.t.

$$B_d(y, r') \subseteq B_d(x, r)$$

Then

$$B_d(y, r') \cap Y \subseteq B_d(x, r) \cap Y$$

□

Example 1.2.17. Let X, Y be nonempty topological spaces. Equip $X \times Y$ with the product topology. For each $y \in Y$, the set $X \times \{y\}$ is a subspace of $X \times Y$, and $X \times \{y\}$ is in bijection with X .

Two topologies on X :

- Original topology

- Subspace topology on $X \times \{y\}$ identified with X .

Proposition 1.2.18. *These two topologies are the same.*

Example 1.2.19. Let X be a set with an order $<$. Let $Y \subseteq X$. Then $<$ gives an order relation of Y .

Two topologies on Y :

- Order topology on Y
- Subspace topology

Proposition 1.2.20. *These do not have to agree!.*

Proof. Let $X = \mathbb{R} \times \mathbb{R}$, with dictionary order and $Y = [0, 1] \times [0, 1]$. Consider $\{1/2\} \times [0, 1]$: This is open in the subspace topology since $\{1/2\} \times (-\epsilon, 1 + \epsilon)$ is an interval in X . This is not open on the order topology since it has the endpoints. $\square$

1.3 Continuous Functions

Definition 1.3.1. Let X, Y be topological spaces. A function $f : X \rightarrow Y$ is continuous if, for every open $U \subseteq Y$, the set $f^{-1}(U)$ is open in X .

Definition 1.3.2. A bijection $f : X \rightarrow Y$ is called a homeomorphism if it is continuous and $f^{-1} : Y \rightarrow X$ is also continuous.

Equivalently: If $f : X \rightarrow Y$ is a bijection, then it induces a bijection $P(X) \rightarrow P(Y)$. If X, Y are topological spaces, a bijection $f : X \rightarrow Y$ is a homeomorphism if it identifies topology on X with the topology on Y .

Example 1.3.3. Let $X = \{a, b\}$ and $Y = \{c, d\}$. Let $\mathcal{T}_X = \{\emptyset, X, \{a\}\}$. Let $\mathcal{T}_Y = \{\emptyset, Y\}$. Consider $f : X \rightarrow Y$ s.t. it is a bijection this is clearly continuous since $f^{-1}(Y) = (X)$. However for $f^{-1} : Y \rightarrow X$ this is not continuous.

Definition 1.3.4. Let $f : X \rightarrow Y$, let $Z \subseteq Y$ be the subspace $f(X)$. If $f : X \rightarrow Z$ is a homeomorphism, then we say that f is a topological embedding.

This is not the same as a continuous injective map.

Example 1.3.5. Let $X = [0, 2\pi)$, $f : X \rightarrow \mathbb{R}^2$ given by $f(t) = (\cos(t), \sin(t))$.

Proposition 1.3.6. Let $f : X \rightarrow Y$ be a map between topological spaces X, Y .

$$f \text{ is continuous} \iff f^{-1}(B) \text{ is open for every } B \text{ in a basis for } Y.$$

Proof. $(\implies)$: immediate by the fact that basis sets are open.

$(\impliedby)$: use fact that every open set in Y is a union of basis sets. □

Definition 1.3.7. Let X be a topological space, $x \in X$. A neighborhood of x is any open set $U \subset X$ containing x .

Definition 1.3.8. A map $f : X \rightarrow Y$ between topological spaces is continuous at $x \in X$ if, for every neighborhood V of $f(x)$, there is a neighborhood U of x s.t. $f(U) \subseteq V$.

Proposition 1.3.9. *Let $f : X \rightarrow Y$ be a map between topological spaces then the following are equivalent.*

(i) f is continuous

(ii) f is continuous at every $x \in X$

(iii) $f^{-1}(C)$ is closed for every closed $C \subseteq Y$

Proof. (i) $\iff$ (iii) : Let $U \subseteq Y$ be open.

Thus, $X \setminus f^{-1}(U) = f^{-1}(Y \setminus U) = f^{-1}(Y \setminus U)$

Thus, $f^{-1}(U)$ is open $\iff f^{-1}(Y \setminus U)$ is closed.

(i) $\implies$ (ii) : Let $x \in X$, let V be a neighborhood of $f(x)$. Then $f^{-1}(V)$ is open, so it's a neighborhood of x and $f(f^{-1}(V)) \subseteq V$.

(ii) $\implies$ (i) : Let $V \subseteq Y$. For each $x \in f^{-1}(V)$, $f(x) \in V$, so there is a neighborhood U_x of x s.t. $f(U_x) \subseteq V$, meaning $U_x \subseteq f^{-1}(V)$.

Thus, $f^{-1}(V) = \bigcup_{x \in f^{-1}(V)} U_x$, so $f^{-1}(V)$ is open. $\square$

Definition 1.3.10. Let X, Y be metric spaces. A function $f : X \rightarrow Y$ is continuous at $x \in X$ if $\forall \epsilon > 0, \exists \delta > 0$ s.t. whenever $d(x, y) < \delta$, then $d(f(x), f(y)) < \epsilon$.

Also the same as, for every ball $B(f(x), \epsilon), \exists \delta$ s.t. $f(B(x, \delta)) \subseteq B(f(x), \epsilon)$.

Proposition 1.3.11. *These definition of continuity are equivalent when X, Y are metric spaces.*

Lemma 1.3.12. *Let $f : X \rightarrow Y, g : Y \rightarrow Z$ be continuous between topological spaces. Then $g \circ f$ is continuous.*

Proof. Let $U \subseteq Z$ be open then $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$. $g^{-1}(U)$ is open, since g is continuous then $f^{-1}(g^{-1}(U))$ is open, since f is continuous. $\square$

Proposition 1.3.13. *Let X be a topological space, $A \subseteq X$ a subspace then the inclusion $f : A \hookrightarrow X$ is continuous.*

Proof. Let $U \subseteq X$ be open, thus $f^{-1}(U) = U \cap A$ is open in A by definition of subspace topology. $\square$

Proposition 1.3.14. *Let $f : X \rightarrow Y$ be a map between topological spaces, suppose that $f(X) \subseteq Z$ for some subspace $Z \subseteq Y$. Then $f : X \rightarrow Z$ is continuous $\iff f : X \rightarrow Y$ is continuous.*

Proposition 1.3.15. *If $f : X \rightarrow Y$ is continuous, and A is a subspace of X then $f|_A : A \rightarrow Y$ is continuous.*

Definition 1.3.16. Let X be a topological space. An open covering of X is a collection $\{U_\alpha\}_{\alpha \in J}$ of open sets s.t. $X = \bigcup_{\alpha \in J} U_\alpha$.

Proposition 1.3.17. Let $f : X \rightarrow Y$, let $\{U_\alpha\}_{\alpha \in J}$ be an open covering. Then f is continuous $\iff$ each restriction $f|_{U_\alpha}$ is continuous.

Proposition 1.3.18. Suppose X is a union of finitely many closed subsets $C_1, \dots, C_n$. Then a map $f : X \rightarrow Y$ is continuous $\iff$ each restriction $f|_{C_i} : C_i \rightarrow Y$ is continuous.